

*2.

Figure 2.1

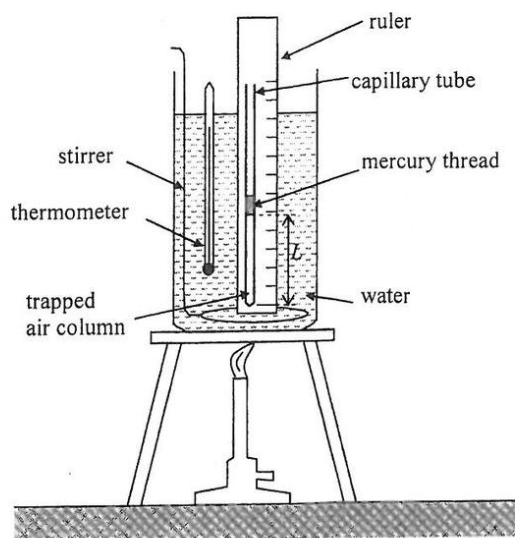


Figure 2.1 shows an air column trapped by a small mercury thread inside a uniform capillary tube. The set-up is heated by a water bath. The length of the air column L is measured at various temperature θ . Some of the results are tabulated below:

Temperature $\theta / ^\circ\text{C}$	20		92
Length of air column L / mm	64		80

- (a) Describe the procedure(s) to be done before taking a reading in order to ensure that the trapped air reaches the same temperature as the water. (2 marks)

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(b) Assume that length L increases linearly with temperature θ throughout.

(i) Estimate the length of the air column when the temperature indicated by the thermometer is 65°C . (2 marks)

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(ii) Find the 'absolute zero' as obtained from this experiment.

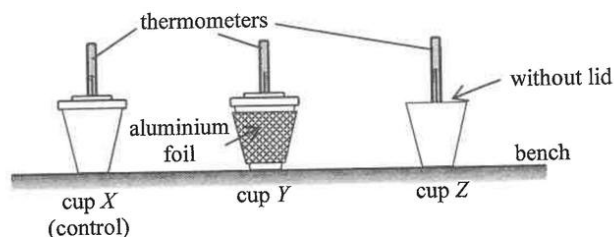
(2 marks)

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Section B: Answer ALL questions. Parts marked with * involve knowledge of the extension component. Write your answers in the spaces provided.

1. A student conducts an experiment to study how to best retain the warmth of water by using identical paper cups *X*, *Y* and *Z* shown in Figure 1.1. Each cup is filled with 250 cm³ of hot water and cup *X* serves as a control.

Figure 1.1



cup	wrapping	with a lid
<i>X</i>	no	yes
<i>Y</i>	aluminium wrapping	yes
<i>Z</i>	no	no

When the water temperature is 90 °C, the student starts taking thermometer readings every minute. Figure 1.2 shows the variation of temperature (*T*) of water in the cups with time (*t*) elapsed.

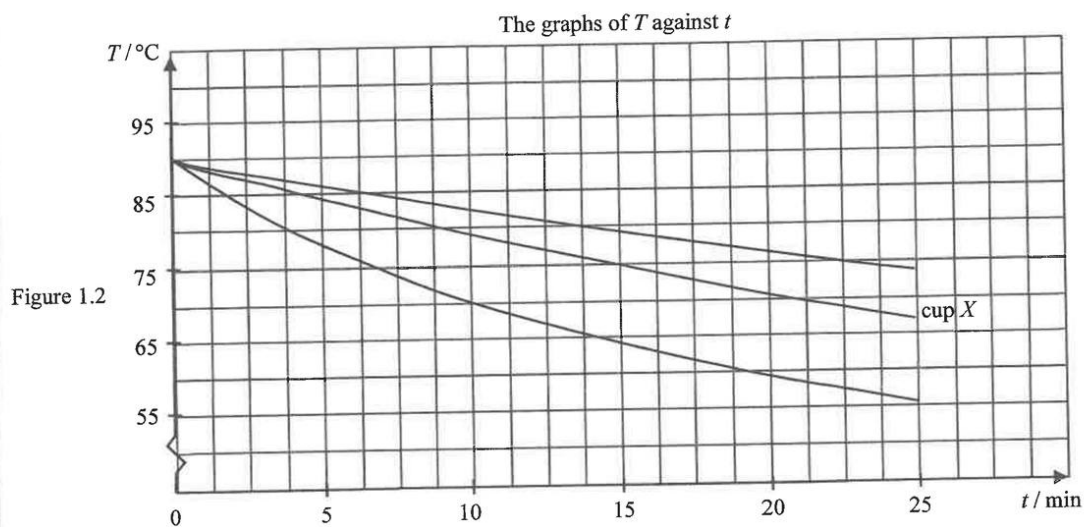


Figure 1.2

- (a) Suggest a reason why this experiment begins with the **same initial water temperature** (90 °C). (1 mark)

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(b) Explain why all the curves become less steep as time elapses.

(2 marks)

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(c) (i) On Figure 1.2, indicate the respective curves that correspond to the results of **cup Y** and **cup Z**. (1 mark)

(ii) Explain your answer by referring respectively to the main heat transfer process. (3 marks)

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(d) Suggest a material for making cups so that heat loss via conduction can be reduced. (1 mark)

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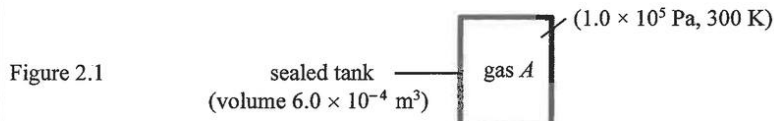
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- *2. (a) (i) Figure 2.1 shows a sealed tank of volume $6.0 \times 10^{-4} \text{ m}^3$ containing a monatomic gas A at a pressure of $1.0 \times 10^5 \text{ Pa}$ and at a temperature of 300 K .



- (I) Estimate the number of gas molecules in the tank, N . (2 marks)

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- (II) Estimate the average kinetic energy of the gas molecules, E_K . (2 marks)

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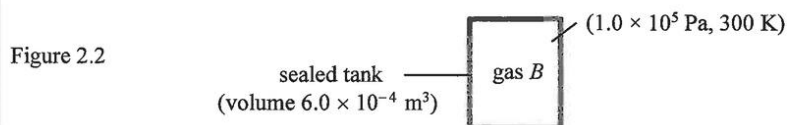
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- (ii) Figure 2.2 shows another identical tank containing monatomic gas B under the same pressure and temperature. A molecule of gas B has $\frac{1}{5}$ the mass of a molecule of gas A .



- (I) State whether N and E_K of gas B are larger than, smaller than or the same as the corresponding values of those of gas A found in (a)(i). (2 marks)

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- (II) Given that the root-mean-square speed ($c_{\text{r.m.s.}}$) of the molecules of gas A is 600 m s^{-1} , estimate $c_{\text{r.m.s.}}$ of the molecules of gas B . (2 marks)

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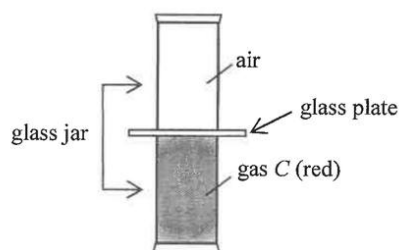
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- (b) Figure 2.3 shows two glass jars containing air and gas C respectively and they are separated by a glass plate. Both are at the same pressure and temperature. Gas C is red in colour.

Figure 2.3



After removing the glass plate, it takes a few minutes for gas C to diffuse several centimetres into the upper jar despite its molecules having a root-mean-square speed of 200 m s^{-1} . Explain this observation. (2 marks)

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Section B: Answer ALL questions. Parts marked with * involve knowledge of the extension component. Write your answers in the spaces provided. Take $g = 9.81 \text{ m s}^{-2}$.

1. A bottle of water of temperature 4°C is taken out from a refrigerator and placed on a table. The room temperature is 30°C .



Figure 1.1

- (a) State one way of heat transfer from the surroundings to the bottle of water. (1 mark)

- (b) Explain whether wrapping the bottle with an aluminum foil will speed up the rate of temperature increase. (2 marks)

- (c) After a period of time, water droplets form on the surface of the bottle.

- (i) State the source of the water droplets. (1 mark)

- (ii) Suppose 0.40 g of water droplets are collected on the surface of the bottle. Estimate the latent heat released from these droplets. (2 marks)

Given: specific latent heat of vaporization of water $= 2.26 \times 10^6 \text{ J kg}^{-1}$

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Figure 9.1

Figure 9.1 shows a microwave oven. Mary wants to estimate the useful output power of the oven. She is provided with the apparatus and material shown in Figure 9.2.

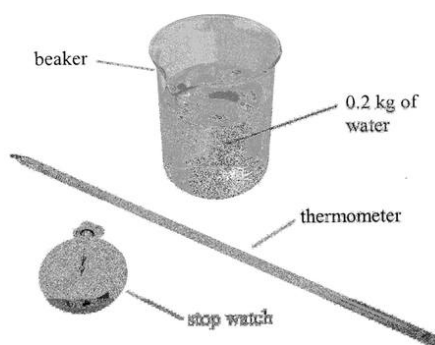


Figure 9.2

- (a) Describe how Mary should conduct the experiment. Specify all measurements Mary has to take. State **EITHER** one precaution taken **OR** one assumption made when conducting this experiment. Write down an equation for calculating the useful output power. (5 marks)

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- (b) The value obtained by Mary is found to be smaller than the rated power of the oven. Suggest one possible reason to account for this difference. (1 mark)

- (c) Explain whether increasing the mass of water used in the experiment would improve the accuracy of the experiment. (1 mark)

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